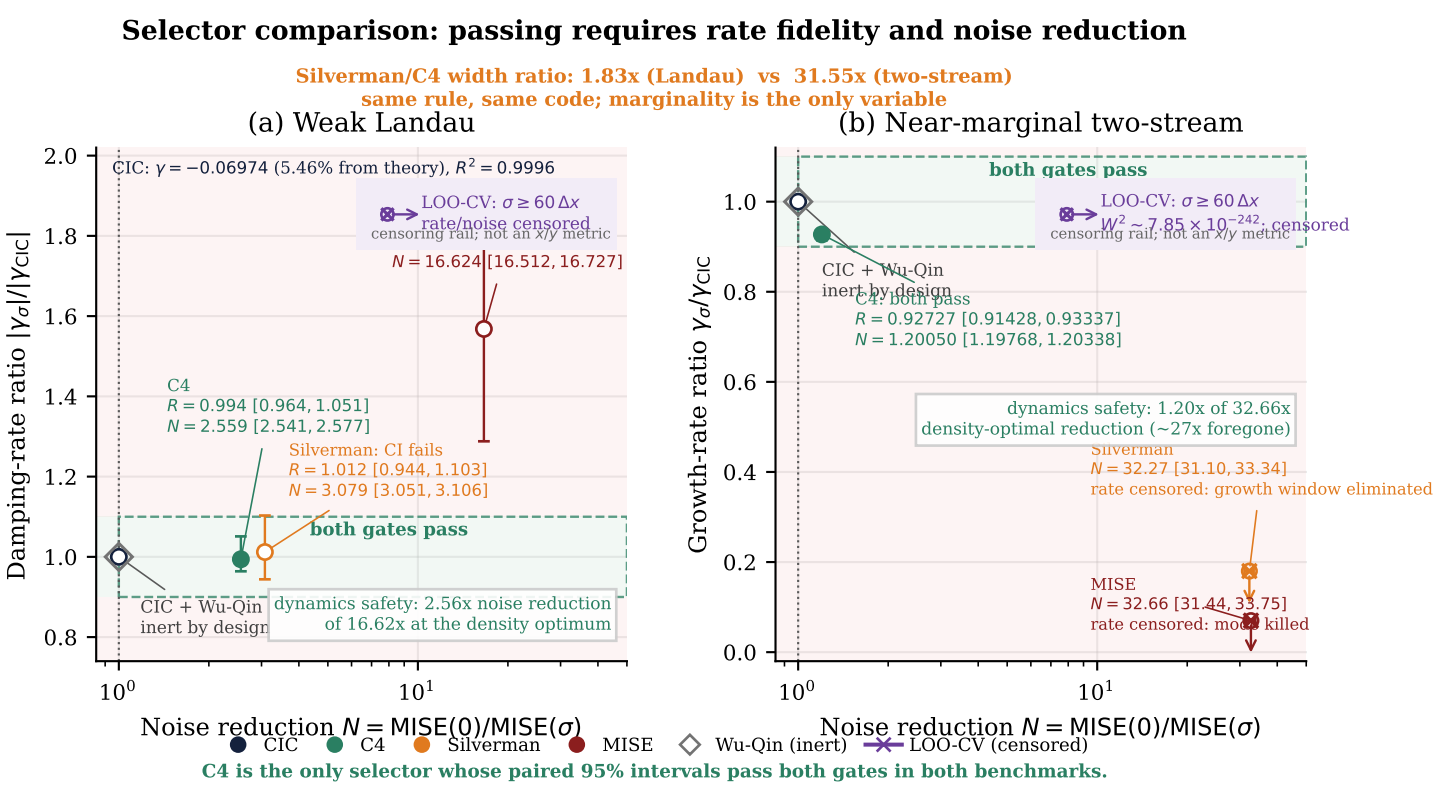

Selector comparison: passing requires rate fidelity and noise reduction
Silverman/C4 width ratio: 1.83x (Landau) vs 31.55x (two-stream)
same rule, same code; marginality is the only variable
(a) Weak Landau
(b) Near-marginal two-stream
Damping-rate ratio $|\gamma_\sigma|/|\gamma_{CIC}|$
Growth-rate ratio $\gamma_\sigma/\gamma_{CIC}$
Noise reduction $N = MISE(0)/MISE(\sigma)$
CIC: $\gamma = -0.06974$ (5.46% from theory), $R^2 = 0.9996$
LOO-CV: $\sigma \geq 60\,\Delta x$
rate/noise censored
censoring rail; not an x/y metric
$N = 16.624$ [16.512, 16.727]
C4
$R = 0.994$ [0.964, 1.051]
$N = 2.559$ [2.541, 2.577]
Silverman: CI fails
$R = 1.012$ [0.944, 1.103]
$N = 3.079$ [3.051, 3.106]
both gates pass
CIC + Wu-Qin
inert by design
dynamics safety: 2.56x noise reduction
of 16.62x at the density optimum
LOO-CV: $\sigma \geq 60\,\Delta x$
$W^2 \sim 7.85 \times 10^{-242}$: censored
CIC + Wu-Qin
inert by design
C4: both pass
$R = 0.92727$ [0.91428, 0.93337]
$N = 1.20050$ [1.19768, 1.20338]
dynamics safety: 1.20x of 32.66x
density-optimal reduction (~27x foregone)
Silverman
$N = 32.27$ [31.10, 33.34]
rate censored: growth window eliminated
MISE
$N = 32.66$ [31.44, 33.75]
rate censored: mode killed
CIC
C4
Silverman
MISE
Wu-Qin (inert)
LOO-CV (censored)
C4 is the only selector whose paired 95% intervals pass both gates in both benchmarks.